Quality Control Layer 2: Instructional Quality Audit

Lesson: Introduction to Machine Learning and Supervised Learning | **Audit Reference:** AUDIT-INST-8C188139

Pedagogical Status: PASS	**Instructional Score:** 93.0%	**Recommendations:** 1

1. Pedagogical Dimensions Assessment

Objective Alignment	Completely aligns with objectives: clearly differentiates supervised from unsupervised learning, and contrasts classification with regression using relatable examples.
Clarity & Scaffolding	Definitions of features (X) and labels (y) are succinct. Avoids overwhelming novice learners with matrix calculus while preserving technical precision.
Lesson Structure & Flow	Pedagogical scaffolding starts with conceptual intuition, transitions into side-by-side Scikit-Learn code, and reinforces learning through interactive categorization.
Example Quality	Housing price prediction (Regression) vs luxury tier classification (Classification) uses the exact same feature vector, creating an intuitive mental model of the difference.
Activity & Hands-on Efficacy	Active problem-classification exercise engages critical thinking by presenting ambiguous real-world scenarios.
Assessment & Knowledge Checks	Questions directly target common misconceptions (e.g., using linear regression on multi-class targets).
Time Allocation (<= 45 mins)	Total duration is 45 minutes, comfortably within the 45-minute constraint.

2. Instructional Weaknesses & Actionable Enhancements (1 item)

ID / Severity	Affected Section & Content	Pedagogical Critique	Actionable Improvement
INST-686CF2 LOW	**Section:** 7. Hands-on Activity: Problem Classification Exercise *"Solutions: 1=Unsupervised Clustering, 2=Supervised Regression, 3=Supervised Classification."*	Displaying solutions immediately adjacent to the problem prompts risks passive reading rather than active recall.	**Recommendation:** Include a collapsible 'Self-Check Reveal' or guided reflection prompt so students attempt categorization before reading solutions.

3. Summary & Educational Impact

Instructional Quality QC Audit PASSED. Strong pedagogical design with effective conceptual contrast.